Water Security Plan Report

Date : 02/ 02/ 2024 Time : 12:40 PM User Name : SPMU_Maharashtra_27_asdf@asdf.com

STATE: MAHARASHTRA_27 » DISTRICT: AHMEDNAGAR_466 » BLOCK: KARJAT_4279 » GRAM PANCHAYAT: AKHONI_167761 » YEAR: 2024-25 »

Gram Panchayat Information

State:	Maharashtra_27	District:	Ahmednagar_466
Block / Taluk :	Karjat_4279	GP Name :	Akhoni_167761
GP ID :	167761	Block Area (Ha) :	150700.0000
GP area (Ha) :	732.0000	Watershed Code :	SRWT005
Watershed name :	ad	Watershed area in Block / Taluk (Ha) :	233
Sub-basin Code :	23	Sub basin name :	Bhima Upper
Basin Code :	233	Basin Name :	Krishna

Location Map

Watershed Map

Aquifer Map

Ground Water Level

S.No.	Site Name	Location Details	Latitude	Longitude	Date	Well ID	Type of Well	Other Well Type	Ownership Type	Government Source	Diameter of well(m)	Depth of Well(mbmp)	Height of Measuring Point (mbgl)	Water Level (mbgl)	Water Level Date

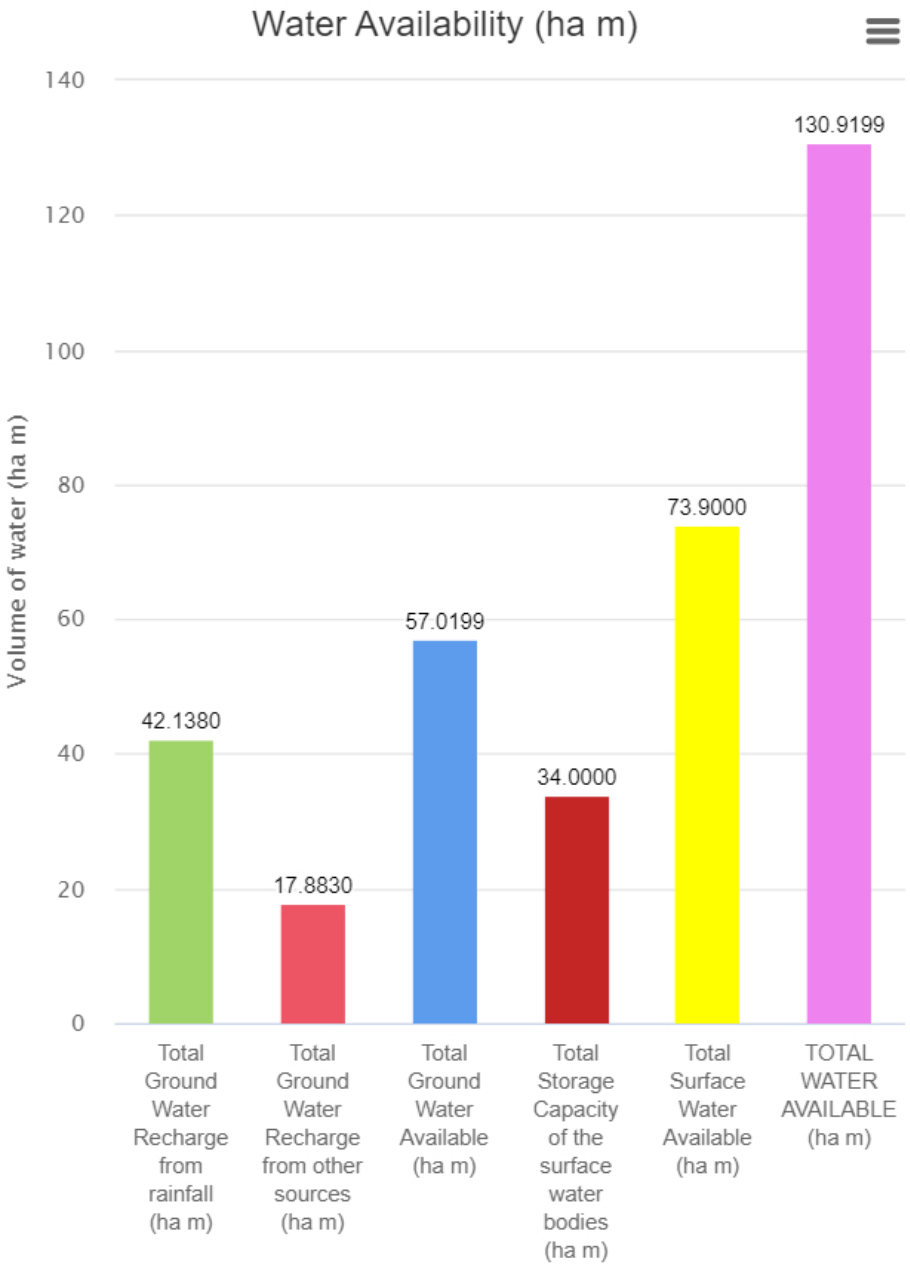
Ground Water Quality

S.No.	Village Name	Location	Latitude	Longitude	Year	Well Type	Source	Well ID

S.No.	Well ID	Well Depth (mbgl)	Aquifer	PH	EC (µS/cm)	TDS (mg/L)	TH (mg/L)	Ca (mg/L)	Mg (mg/L)	Na (mg/L)	K (mg/L)	Carbo nate (mg/L)	Bicar bonate (mg/L)	Sulph ate (mg/L)	Chlor ide (mg/L)	Fluo ride (mg/L)	Nitr ate (mg/L)

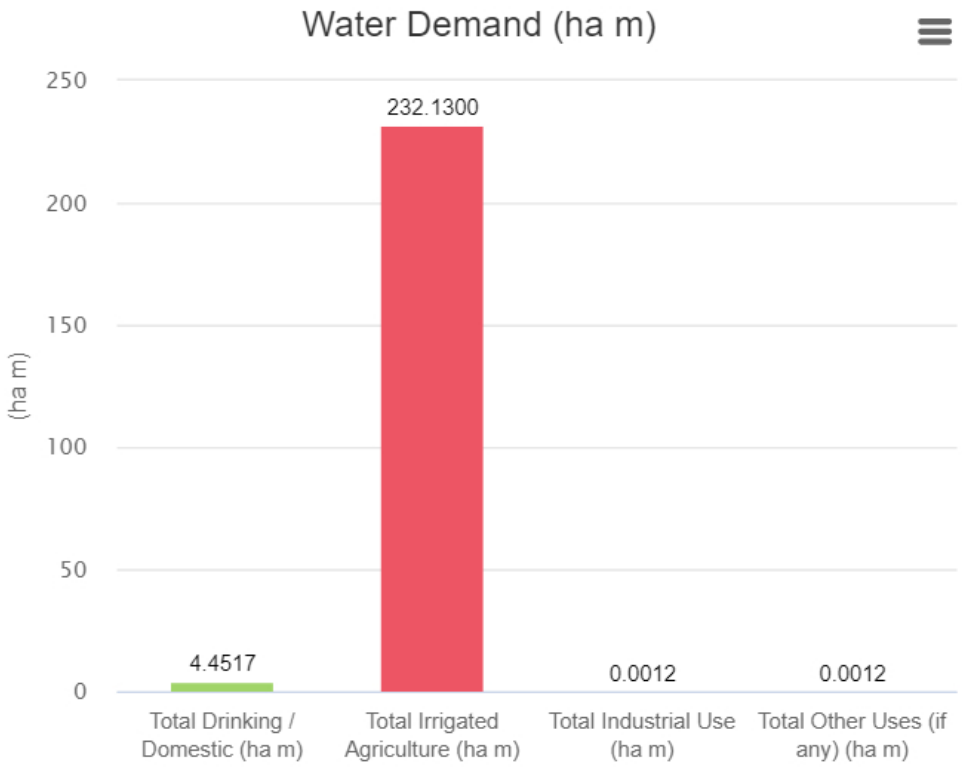
Water Availability Report

Water Availability		
1.	Total Available Water From Rainfall	
a.	Annual Rainfall (mm)	528.7400
b.	Estimated Volume of Rainfall (ha m)	387.0377
c.	Evapotranspiration rate (mm)	291.3700
d.	Estimated Evapotranspiration losses (ha m)	213.2828
e.	Estimated Water Available From Rainfall (ha m)	173.7549
f.	Weighted Average Runoff Coefficient	0.3000
g.	Rainfall Runoff(ha m)	52.1265
2.	Ground Water	
a.	Recharge From Rainfall (ha m)	
(i)	Monsoon	34.5224
(ii)	Non-Monsoon	7.6156
	Total Ground Water Recharge from rainfall (ha m)	42.1380
b.	Ground Water Recharge from other sources (ha m)	
(i)	Monsoon	4.9640
(ii)	Non-Monsoon	12.9190
	Total Ground Water Recharge from other sources (ha m)	17.8830
c.	Losses Due to Natural Discharge (ha m)	3.0011
	Total Ground Water Available (ha m)	57.0199
d.	Ground Water Withdrawal (ha m)	36.0349
3.	Surface Water	
a.	Surface Water available in storages	
(i)	Number of Surface Water Bodies and Storage Capacity (ha m)	
Type of Structures	No. of Structures	Storage Capacity (ha m)
Ponds	45	34
	Total Storage Capacity of the surface water bodies (ha m)	34.0000
(ii)	Losses if Any (%)	15.0000
(iii)	Surface Water Available in Storages (ha m)	28.9000
b.	Transported Water from outside GP area (ha m)	45.0000
c.	Total Surface Water Available (ha m)	73.9000
4.	TOTAL WATER AVAILABLE	
	TOTAL WATER AVAILABLE (ha m)	130.9199



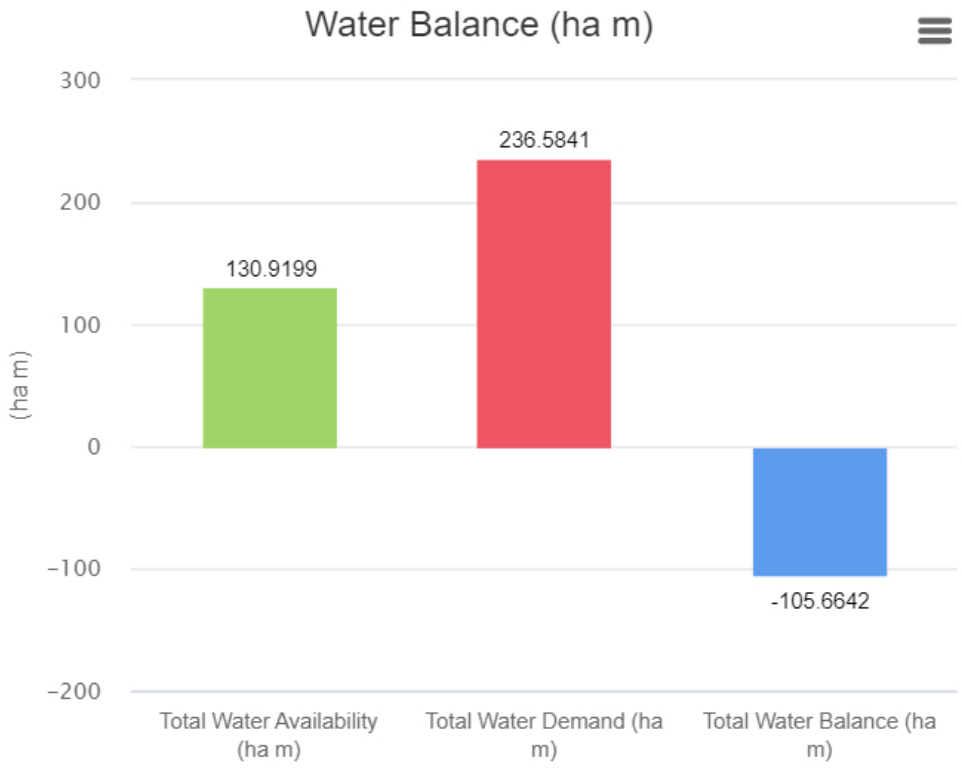
Water Demand Report

Water Demand		
A	Drinking /Domestic (ha m)	
i)	From Surface Water	0.0000
ii)	From Ground Water	4.4517
	Total Drinking /Domestic (ha m)	4.4517
B	Irrigated Agriculture (ha m)	
i)	From Surface Water	11.7000
ii)	From Ground Water	220.4300
	Total Irrigated Agriculture (ha m)	232.1300
C	Industrial Use (ha m)	
i)	From Surface Water	0.0007
ii)	From Ground Water	0.0005
	Total Industrial Use (ha m)	0.0012
D	Other Uses (if any) (ha m)	
i)	From Surface Water	0.0005
ii)	From Ground Water	0.0007
	Total Other Uses (if any) (ha m)	0.0012



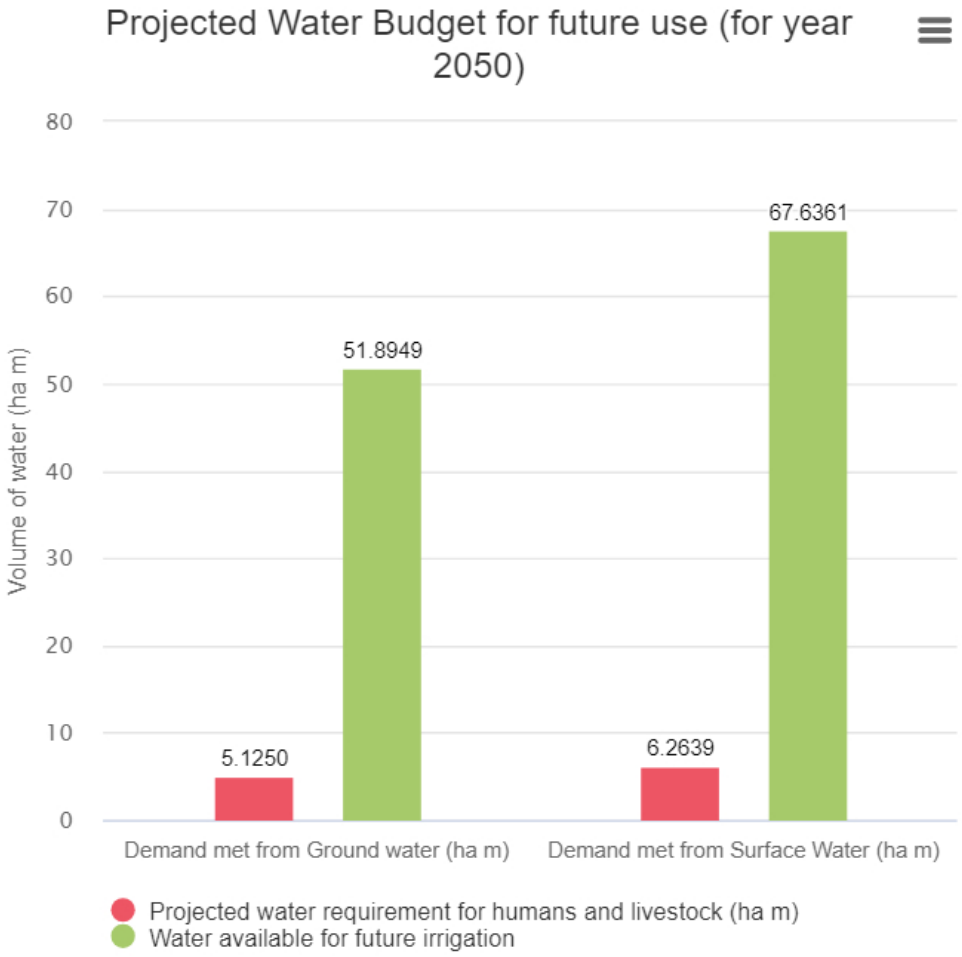
Water Balance Report

Water Balance		
A	Water Availability (ha m)	
i)	Total Ground Water Available	57.0199
ii)	Total Surface Water Available	73.9000
	Total Water Availability (ha m)	130.9199
B	Water Demand (ha m)	
i)	Total Surface Water Demand	11.7012
ii)	Total Ground Water Demand	224.8829
	Total Water Demand	236.5841
C	Balance [Surplus (+) / Deficit (-)]	
i)	Total Ground Water Balance	-167.8630
ii)	Total Surface Water Balance	62.1988
iii)	Total Water Balance	-105.6642



Water Budget Report

A	Allocation for Drinking & Domestic Requirements	
I	Humans	
i)	Present Human Population	1560.0000
ii)	Annual population growth rate (r)	4.0000
iii)	No. of years	27.0000
iv)	Projected human population (as in 2050)	4678.0000
v)	Per capita requirement (L)	55.0000
vi)	Projected water requirement for humans (as in 2050) (ha m)	9.3911
II	Livestock	
i)	Present livestock population	1539.0000
ii)	Annual population growth rate (r)	1.5000
iii)	No. of years	27.0000
iv)	Projected livestock population (as in 2050)	2335.0000
v)	Average per capita requirement (L)	23.4405
vi)	Projected water requirement for livestock (as in 2050) (ha m)	1.9978
B	Projected water requirement for humans and livestock (ha m) (as in 2050)	
	Total Projected water requirement for humans and livestock (as in 2050) (ha m)	11.3889
i)	Proposed share to be met from Ground water (in %)	45.0000
ii)	Proposed share to be met from Surface water (in %)	55.0000
iii)	Demand met from Ground water (ha m)	5.1250
iv)	Demand met from Surface water (ha m)	6.2639
C	Water available for future irrigation	
i)	Ground Water	51.8949
ii)	Surface water	67.6361
iii)	Total water available for future irrigation	119.5310

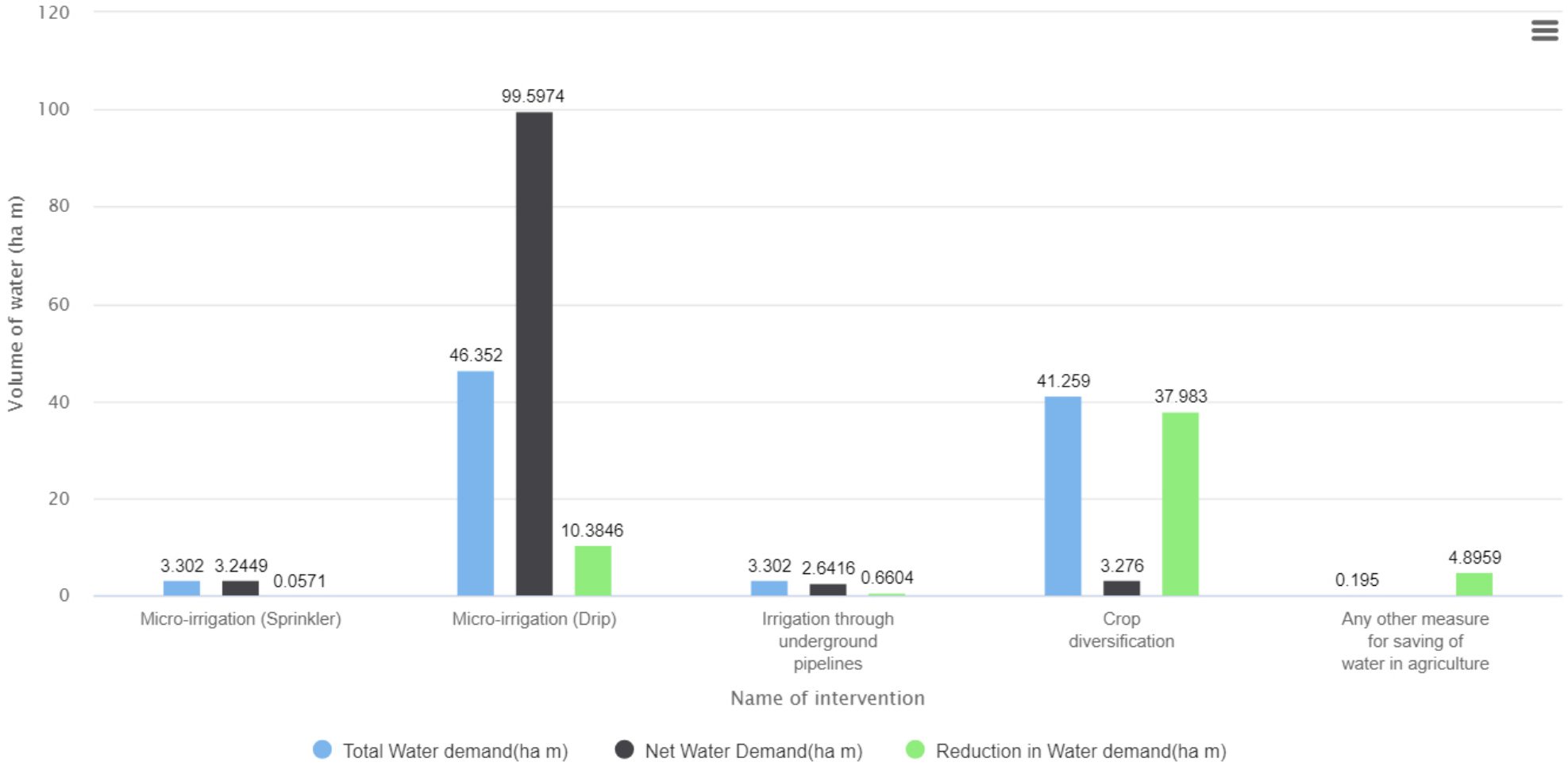


Demand Side

Total Area under Conventional Irrigation (ha) : 1432

Demand Side						
Sr No	Name of Intervention	Proposed Area(ha)	Total Water demand(ha m)	Net Water Demand(ha m)	Reduction in Water demand(ha m)	Approximate cost(in lakh)
1	Sprinkler	3.0000	3.3020	3.2449	0.0571	0.0000
2	Drip	109.0000	46.3520	99.5974	10.3846	67.2000
3	Pipelines	52.0000	3.3020	2.6416	0.6604	45.0000
4	Crop Diversification	52	41.2590	3.2760	37.9830	0.0000
5	Other Measures	45.4000	0.1950	--	4.8959	59.0000
	Total	261.4000	94.4100	108.7599	53.9810	171.2000

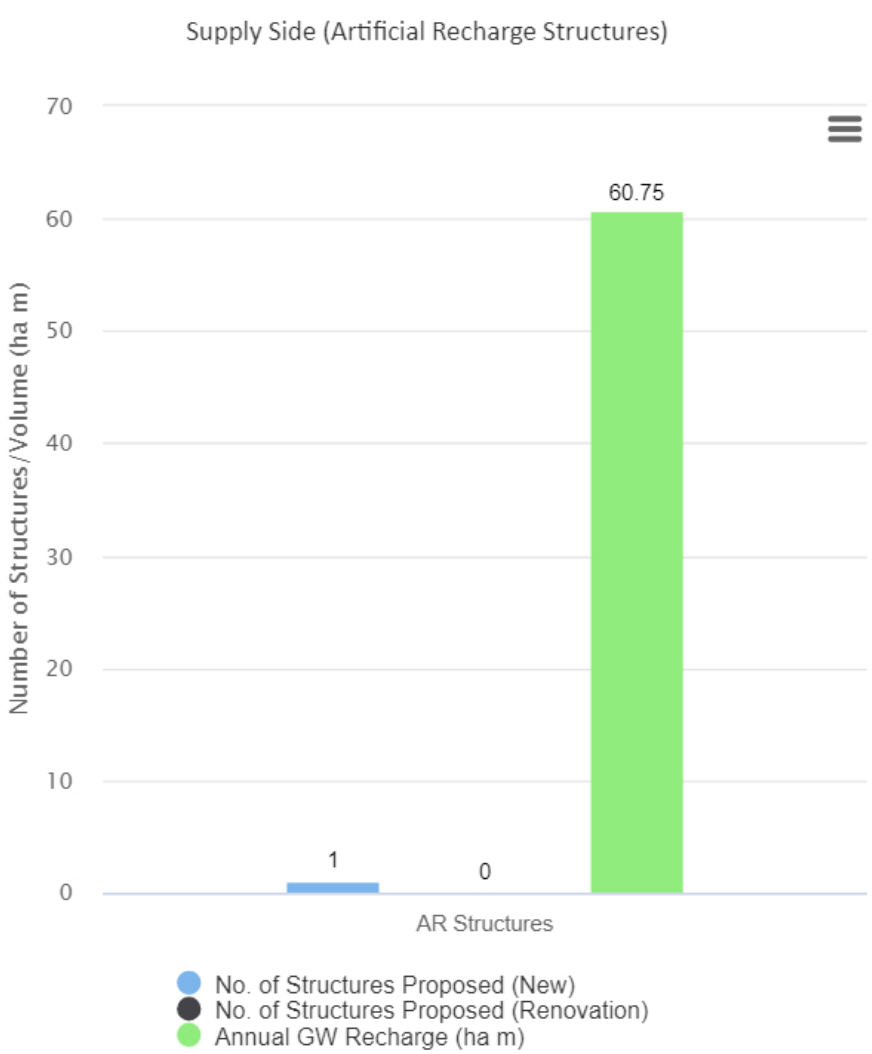
Funding Source						
Sr No	Name of Intervention	Area Proposed (ha)	Financial Year	Department Name	Scheme Name	Approximate Cost(in Lakh)
1	Drip	60	2023-24	Department of Agriculture	PMKSY(Per Drop More Crop)	67.2
2	UG Pipelines	52	2024-25	Gujarat Green Revolution Company Ltd. (GGRC)	Gray Water Management-Phase-II	45
3	Other Measures	0.2	2023-24	Department of Agriculture	MIDH (Mission for Integrated Development of Horticulture)	5
4	Other Measures	0.2	2023-24	Department of Agriculture	MIDH (Mission for Integrated Development of Horticulture)	20
5	Other Measures	45	2024-25	Gujarat Water Resources Development Corporation Ltd. (GWRDC)	Jal Jeevan Mission	34
	Total	157.4				171.2000

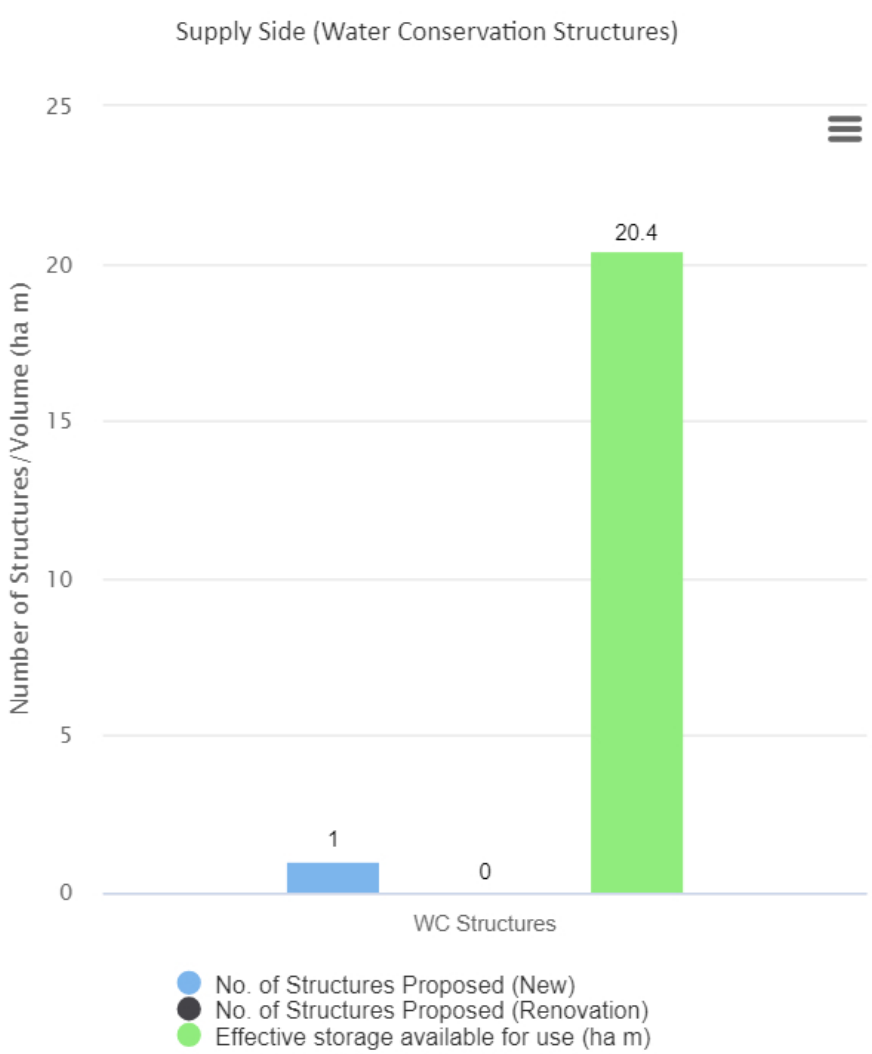

120
100
80
60
40
20
0
Volume of water (ha m)
99.5974
46.352
41.259
37.983
10.3846
4.8959
3.302
3.2449
0.0571
3.302
2.6416
0.6604
3.276
0.195
Micro-irrigation (Sprinkler)
Micro-irrigation (Drip)
Irrigation through underground pipelines
Crop diversification
Any other measure for saving of water in agriculture
Name of intervention
Total Water demand(ha m)
Net Water Demand(ha m)
Reduction in Water demand(ha m)

Supply Side

Supply Side (Artificial Recharge Structures)										
SR No	Name of Structure	Storage Capacity(ha m)	Annual GW Recharge(ha m)	Village Name	Latitude	Longitude	Financial Year	Department Name	Scheme Name	Approximate Cost (in Lakhs)
1	Gully Plugs(New)	45	60.75	Akhoni	34.454	44.54	2024-25	Department of Forest	BVPY Pulse	34
	Total	45.0000	60.7500							34.0000

Supply Side (Water Conservation Structures)										
SR No	Name of Structure	Storage Capacity(ha m)	Effective storage available for use (ha m)	Village Name	Latitude	Longitude	Financial Year	Department Name	Scheme Name	Approximate Cost (in Lakhs)
1	Farm Pond(New)	34	20.4	Akhoni	45.	45.3	2024-25	Department of Fisheries	Lift Irrigation Scheme (LIS)	34
	Total	34.0000	20.4000							34.0000


Supply Side (Artificial Recharge Structures)
70
60
50
40
30
20
10
0
Number of Structures/Volume (ha m)
60.75
1
0
AR Structures
No. of Structures Proposed (New)
No. of Structures Proposed (Renovation)
Annual GW Recharge (ha m)


Supply Side (Water Conservation Structures)
25
20
15
10
5
0
Number of Structures/Volume (ha m)
20.4
1
0
WC Structures
No. of Structures Proposed (New)
No. of Structures Proposed (Renovation)
Effective storage available for use (ha m)

PGWM Committee

PGWM Committee (VWSC/WUA/CBO)				
Sr.no	Name	Designation	Gender	Mobile /Landline No.
1	AAS	ass	Male	4544545445

Details of PGWM Meeting in which resolution was passed												
Sr.no	State	District	Block	Gram Panchayat	Date	GP Name	Village Name	Meeting Location	Male	Female	Transgender	Total number of attendees
1	Maharashtra_27	Ahmednagar_466	Karjat_4279	Akhoni_167761	2024-01-31	Akhoni_167761	Akhoni	43	34	23	34	91

Photograph


Govt. of India
Ministry of Jal Shakti
Dept. of Water Resources, River Development
& Ganga Rejuvenation
आज़ादी का
अमृत महोत्सव
Atal Bhujal Yojana
G20